



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

By Teodora Ivanova
August 2026



Outline

- Executive Summary
 - Project Overview & Business Impact
- Introduction
 - The Commercial Space Race & Landing Economics
- Methodology
 - Data collection, Wrangling, EDA, Interactive Analytics & ML
- Results
 - Insights & Model Comparisons
- Conclusion
 - Key takeaways & Insights
- Appendix

Executive Summary: Methodologies

- **Data Collection & Extraction**
 - Data extracted from the **SpaceX Rest API**
 - Collected Falcon 9 historical launch records from a Wikipedia using **BeautifulSoup**
- **Data Wrangling**
 - Applied mean payload imputation, created binary target variable Class, and performed **One-Hot Encoding**
- **EDA & Visualizations**
 - Evaluated spatial and categorical patterns using **SQL queries**, Seaborn statistical plots, **Folium geospatial maps**, and an interactive **Plotly Dash application**
- **Machine Learning Pipeline:**
 - Standardized features with **StandardScaler** and trained four classification algorithms (**Logistic Regression**, **Support Vector Machines**, **Decision Trees**, and **K-Nearest Neighbors**) using **GridSearchCV** for hyperparameter tuning

Executive Summary: Results

- **Exploratory Insights:**

- **Payload & Orbit Relations:** LEO, ISS, and PO orbits yield higher landing success rates for heavy payloads, whereas for GTO launches both outcomes are present.
- **Site Performance:** Landing success rate improved across all launch sites over time as flight numbers increased.

- **Geospatial Proximity:**

- Launch sites are strategically located near coastlines for safety and transport infrastructure to ease heavy rocket logistics.

- **Predictive Model Performance:**

- All four optimized models achieved **83.33% accuracy score** on the test set.
- **K-Nearest Neighbors (KNN)** performed exceptionally well on cross-validation, demonstrating strong classification accuracy on landing outcomes.

- **Business Impact:**

- Accurately predicting first-stage reusability enables commercial bidders to estimate launch costs near **\$62 Million** versus competitor costs exceeding **\$165 Million**.

Introduction: Project Background & Context

- **SpaceX Claims:**

- **SpaceX** advertises **Falcon 9** rocket launches on its website with a cost of **62 million** dollars, while other providers cost upward of **165 million** dollars each

- **The Why – Reusability:**

- Up to 60–70% of the total manufacturing cost of a Falcon 9 rocket is from the **first stage booster**
- Landing the first stage safely allows SpaceX to reuse the booster, driving down operational cost

- **The Analytical Opportunity:**

- Historical data such as payload specs, landing site and orbital targets contain valuable patterns that determine whether a booster can be safely recovered giving an estimate of the launch cost

Introduction: Problems & Answers

- **Predicting Landing Outcome:**

- *Can we reliably predict whether a Falcon 9 first stage will land successfully ($Class = 1$) or fail ($Class = 0$) using ML?*

- **Payload & Orbital Impact:**

- *How do payload mass and target orbit impact the likelihood of a successful booster landing?*

- **Launch Site Choice:**

- *Do specific launch sites exhibit higher success rates over time?*

- **Cost & Bidding Advantage:**

- *How can a commercial competitor use these predictions to calculate SpaceX's true launch cost and place competitive bids for launch contracts?*

Section 1

Methodology

Methodology

Executive Summary

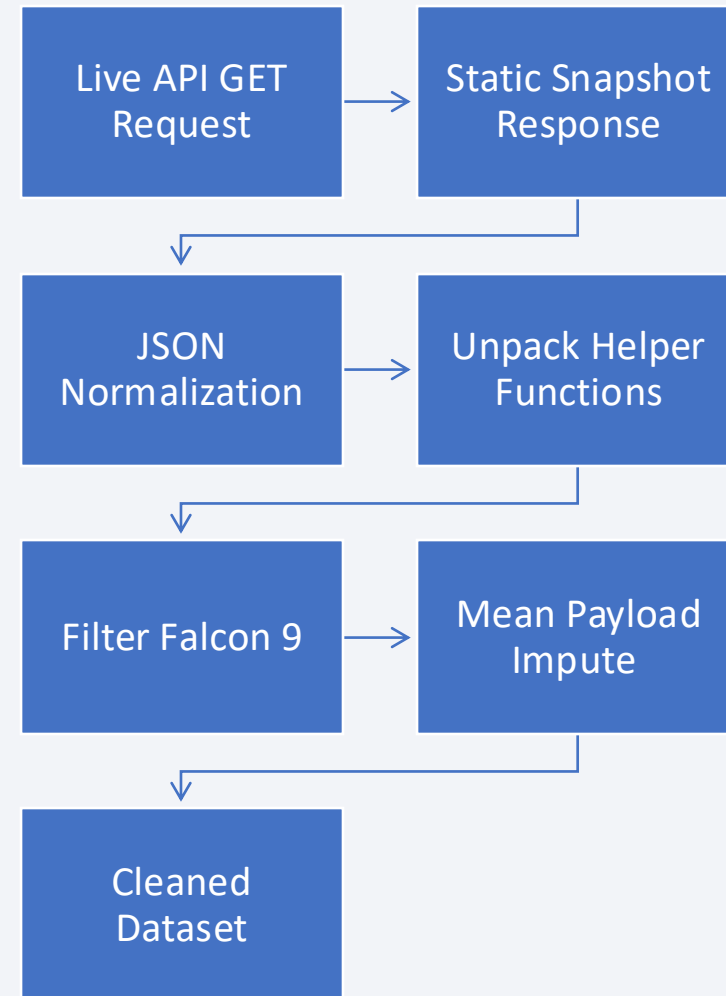
- **Data collection methodology:**
 - Extracted real-time launch records from the **SpaceX REST API**
 - Web-scraped Wikipedia historical launch data using **BeautifulSoup**.
- **Perform data wrangling**
 - Handled missing values using **mean imputation**
 - Created the binary target column: **Class** (1 = Success, 0 = Fail).
 - One-Hot Encoded categorical attributes.
- **Perform exploratory data analysis (EDA) using visualization and SQL**
- **Perform interactive visual analytics using Folium and Plotly Dash**
- **Perform predictive analysis using classification models**
 - How to build, tune, evaluate classification models

Data Collection: SpaceX REST API

- Sent **HTTP GET** requests to the official **SpaceX REST API** (/v4/launches/past) to fetch real-time past launch data.
- Requested the SpaceX launch data with **HTTP GET** request using static response object to keep results more consistent
- Converted the raw JSON response into a **Pandas DataFrame** using **.json_normalize()**.
- Extracted core flight metrics and launch parameters (payload mass, core data, booster version, and launch site) from the **SpaceX REST API** via custom helper functions(**getBoosterVersion()**, **getCoreData()**, **getPayload()**, **getLaunchSite()**).
- Filtered the dataset strictly for **Falcon 9 launches** (excluding Falcon 1 flights).
- Dealt with missing values for the **PayloadMass** column using mean imputation

Data Collection – SpaceX API

- The whole data collection process could be found at [Data Collection SpaceX API](#)

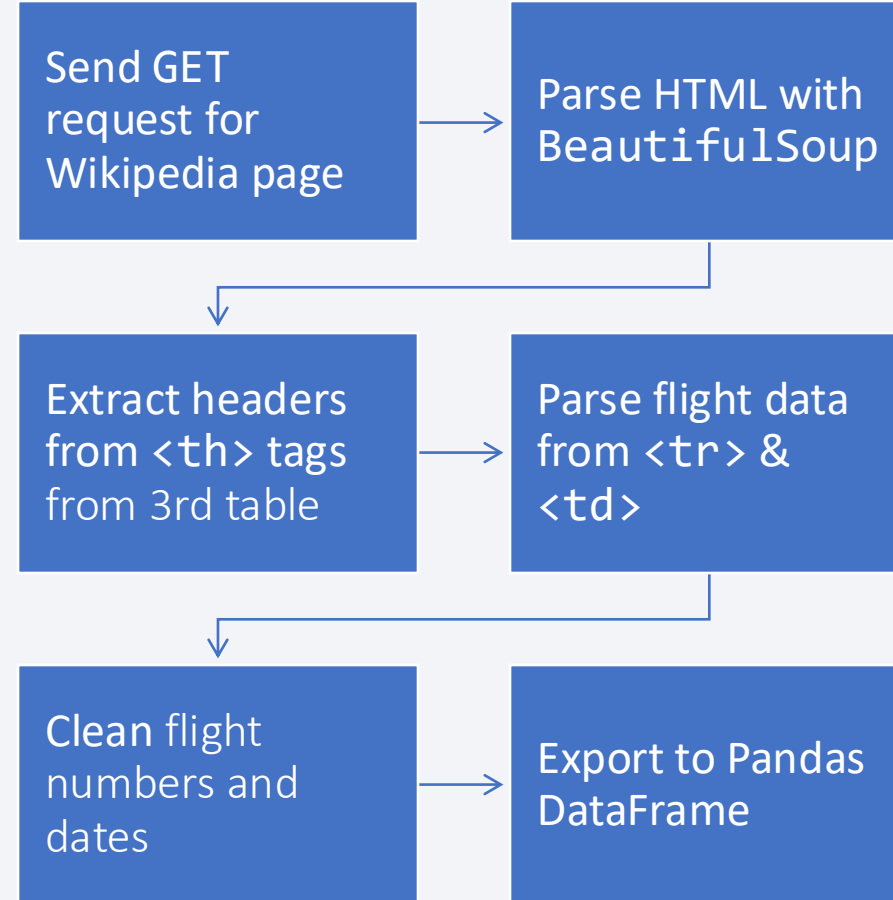


Data Collection: Wikipedia Web Scraping

- Executed an **HTTP GET** request using Python **requests** to fetch the Falcon 9 Launch Wikipedia page
- Used **BeautifulSoup** to parse the raw HTML response and build the DOM tree
- Located the third table and iterated over its **<th>** elements to extract the column names of our dataset
- Built the final DataFrame by iterating through the **<tr>** rows and **<td>** cells of the launch tables to parse flight numbers, dates, booster versions, payload masses, orbits, customers, and landing outcomes.

Data Collection - Scraping

- The whole process can be found at [Web scraping Notebook](#)



Data Wrangling

- **Missing Value Imputation:**

- Handled missing payload mass values in PayloadMass by calculating the mean of known payload weights and performing **mean imputation**.

- **Target Binary Variable Creation (Class):**

- Evaluated raw landing outcomes
- Created binary target variable **Class**:
 - **1 (Successful Landing)**: Returned to Ground Pad (True RTLS) or Drone Ship (True ASDS) or specific region of the Ocean(True Ocean)
 - **0 (Unsuccessful Landing)**: Unsuccessful ocean landings, missed pads, or drone ships ('False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None').

Data Wrangling

- **Categorical Feature Encoding**

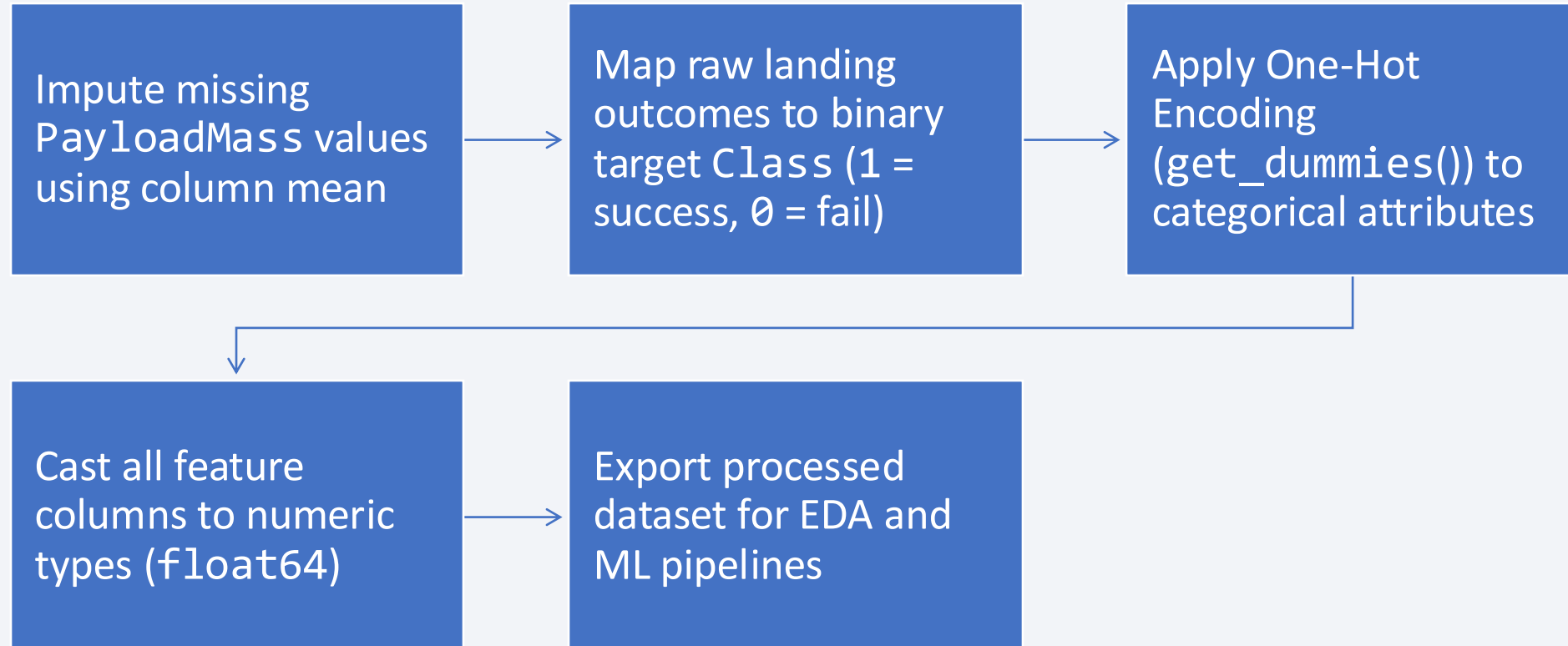
- Applied **One-Hot Encoding** (`pandas.get_dummies()`) to convert categorical features (Orbit, LaunchSite, LandingPad, Serial) into numeric dummy variables for machine learning models.

- **Data Type Conversion:**

- Cast all numeric columns to float64

Notebook with the Data Wrangling is at [Data Wrangling](#)

Data Wrangling



EDA with Data Visualization

- **Key Visualizations & Reasons:**

- **Scatter Plots (catplot):** Plotted *Flight Number vs. Launch Site* and *Payload Mass vs. Launch Site* to visualize how landing success evolved over time across launch locations.
- **Bar Charts:** Calculated and plotted mean success rates by **Orbit Type** to find high-yield and risky orbits.
- **Scatter Plots (scatterplot):** Plotted *Flight Number vs Orbit* to see if success for each Orbit is related to number of flights and *Payload Mass vs. Orbit* to see if successful heavy payload landings are connected to orbit type
- **Line Charts:** Evaluated launch success trends over time (by year) to get the average launch success trend

All Plots can be found at :[EDA Notebook](#)

EDA with SQL

- **SQL Queries Performed:**

- **Unique Launch Sites:** Identified all unique operational launch locations.
- **Site Filtering :** Filtered 5 records starting with "CCA".
- **Payload Calculations:** Used aggregate functions (SUM(), AVG()) to compute total payload mass carried for NASA missions and average payload mass carried by booster version F9 v1.1
- **Date-Range Filtering:** Found the minimum date recorded among successful ground pad landings and ranked the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order.
- **Booster Performance Metrics:** Found the names of the boosters which have successfully landed on a drone ship and have payload mass between 4000 and 6000 and the boosters who carried the maximum payload mass
- **Mission Outcome:** Used **GROUP BY** to find the total number of mission success and failure.
- **2015 drone failure:** Found in which months and which booster failed to land on a drone in 2015

All queries can be found at [EDA with SQL Notebook](#)

Build an Interactive Map with Folium

- **Map Objects Created & Added:**
 - **Location Markers (folium.Marker, folium.Circle):** Added Location Marker for every launch site and NASA. So we can gauge proximity to important targets like Equator, coastline and so on.
 - **Marker Clusters (folium.plugins.MarkerCluster):** Grouped dense launch records into interactive clusters with custom color-coded markers (Green for success Class=1, Red for failure Class=0) to easily identify sites with higher success rates.
 - **Mouse Position(MousePosition):** used mouse-position tool to fetch precise coordinates for interesting targets in the site proximity.
 - **Distance Lines (folium.PolyLine):** Calculated and plotted distance lines from launch sites to nearby infrastructure (coastlines, highways, railways, and cities) to evaluate operational convenience and safety risks.
- All generated maps could be found at : [Folium Site Location](#)

Build a Dashboard with Plotly Dash

- **Dashboard Components & Visualizations**

- **Launch Site Dropdown Filter (dcc.Dropdown):** Allows users to select "All Sites" or isolate a specific launch pad(CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40). This way they can compare performance across pads and compare them to the overall averages.
- **Payload Range Slider (dcc.RangeSlider):** Enables dynamic filtering of payload mass in kilograms (0 kg to 10,000 kg with 1,000 kg intervals). Help us examine how payload weight influence mission success.
- **Success Rate Pie Chart (dcc.Graph):** Renders the proportion of successful vs. failed landings overall, or the total count of successful landings for all sites or a selected launch site. Provides insights at a glance about the percentage breakdown of successful vs. failed landings for the chosen parameters.
- **Payload vs. Outcome Scatter Plot (dcc.Graph):** Displays individual launches mapping Payload Mass vs. Class, color-coded by Booster Version Category (e.g., v1.0, v1.1, FT, B5) again for a chosen site or for all sites. Visualizes how different booster versions handle varying payload weights.

All code used for the dashboard is at :[Dash Application SpaceX](#)

Predictive Analysis (Classification)

- **Data Preparation & Feature Scaling:**
 - Created target vector Y (Class column) and extracted feature matrix X.
 - Standardized feature attributes using StandardScaler to ensure zero mean and standard deviation of one to help distribution sensitive models like the SVM.
- **Train / Test Dataset Split:** Partitioned data into 80% training and 20% testing sets using `train_test_split()`.

Predictive Analysis (Classification)

Hyperparameter Tuning & Cross-Validation:

Used **GridSearchCV** with 10-fold cross-validation to tune hyperparameters for four classification algorithms:

- **Logistic Regression:** Tuned regularization parameters (C, penalty terms, solvers).
- **Support Vector Machine (SVM):** Evaluated kernel functions ('linear', 'rbf', 'poly', 'rbf', 'sigmoid') and margin constraints (C, gamma).
- **Decision Tree Classifier:** Tested splitting criteria (gini, entropy), tree depth (max_depth), and leaf node parameters (splitter, min/max_samples_leaf, max_features).
- **K-Nearest Neighbors (KNN):** Optimized neighbor count, distance metrics (p), and weighting algorithms.

Predictive Analysis (Classification)

Model Evaluation & Best Model Selection:

Evaluated each model using test accuracy scores and confusion matrices to analyze false positive/negative rates.

Result: All four algorithms achieved test accuracy of **83.33%** on the test set, demonstrating strong predictive ability for launch recovery.

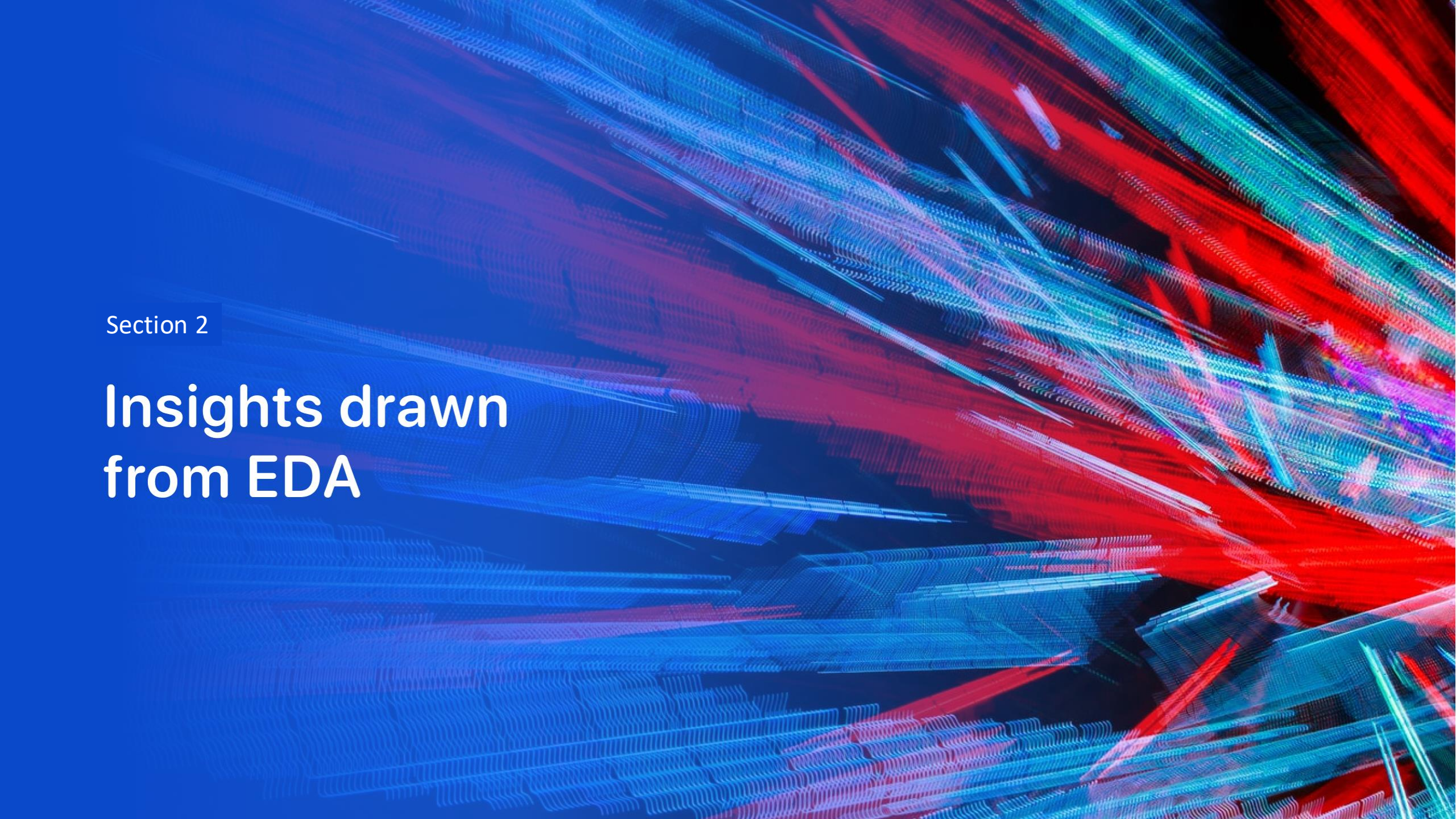
The whole ML flow and results could be found at: [ML_Predictions](#)

Predictive Analysis (Classification): Flowchart



Results

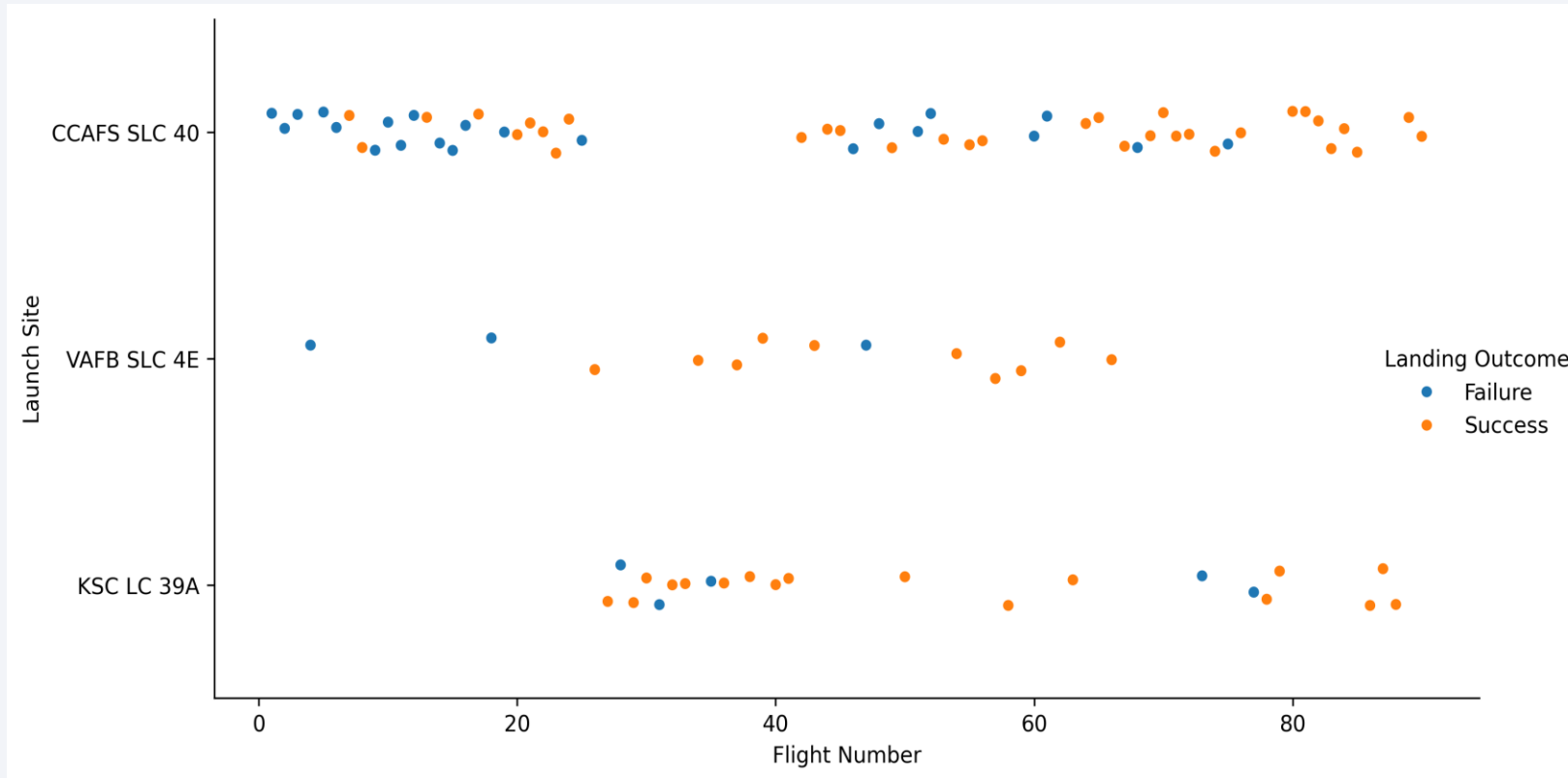
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

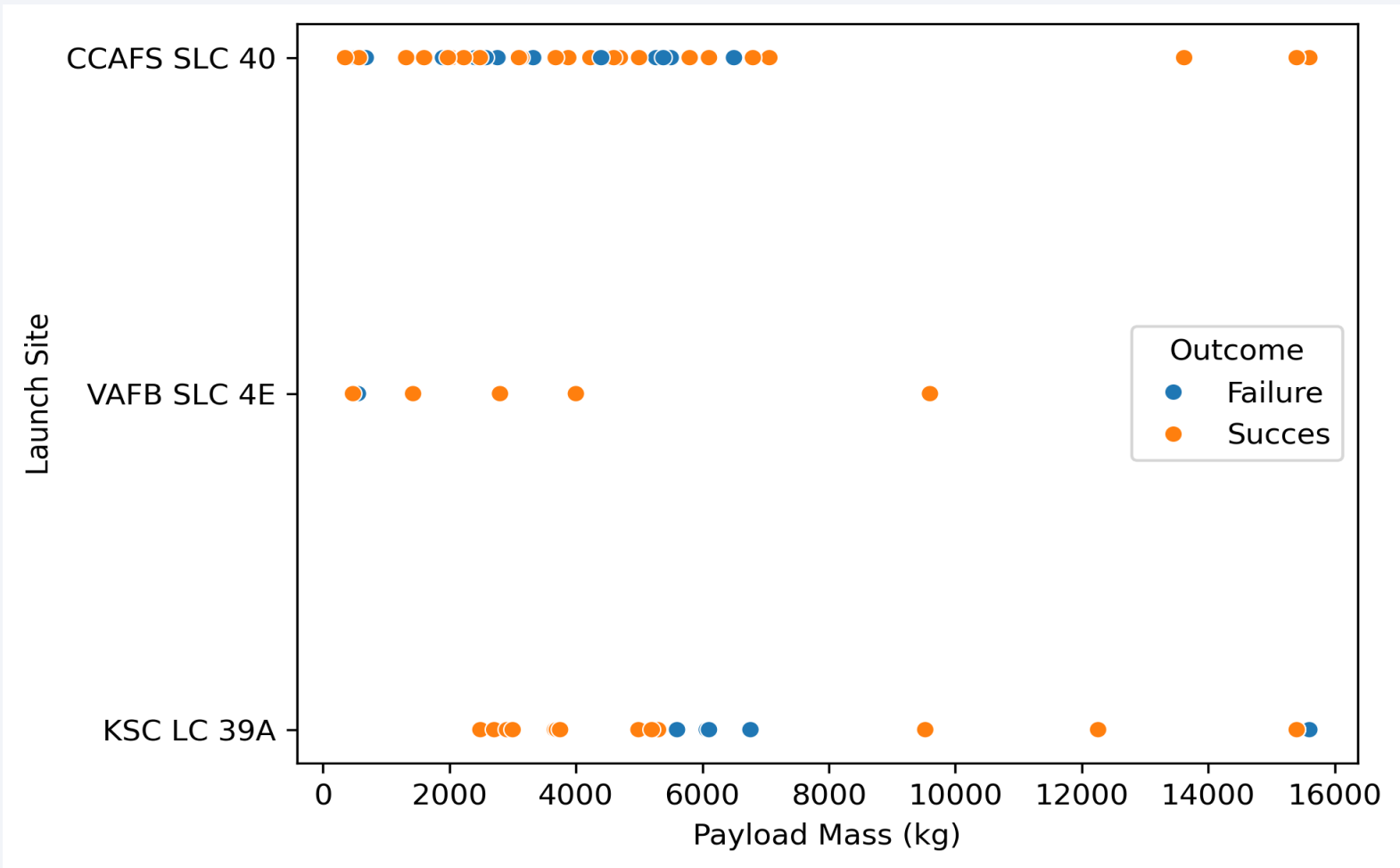
Insights drawn from EDA

Flight Number vs. Launch Site



- We can see that with the increase of the number of flights (>40) the success rate improved significantly. This was probably due to launch technology, operational algorithms and methodology improvement.

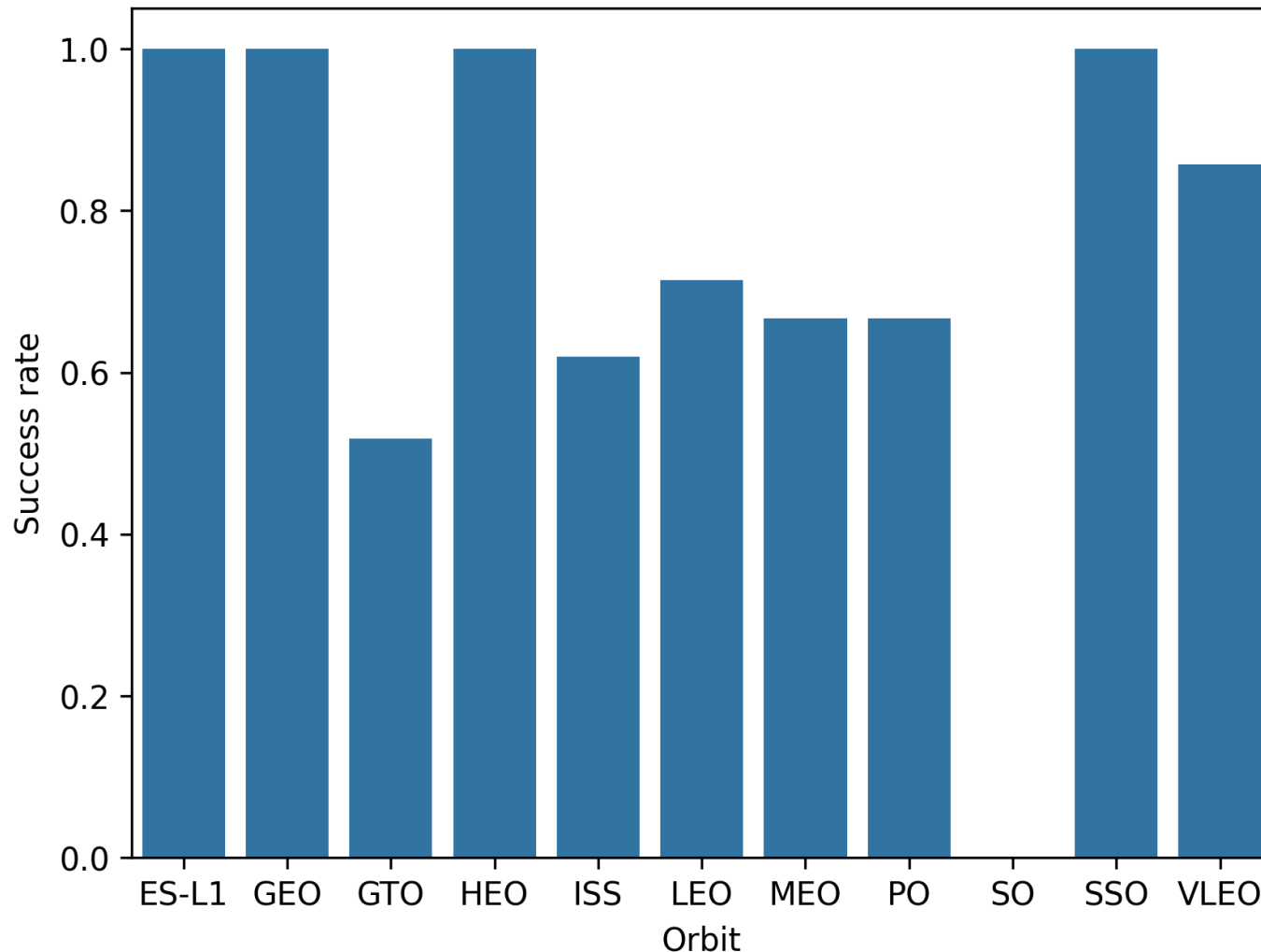
Payload vs. Launch Site



- **VAFB-SLC 4E:** There are no heavier than 10000 kg payloads launched. The success rate for payloads with weight between 1800 and 10000 was 100%.
- **CCAFS SLC 40:** The success rate for mass between 0 and 8000 is pretty mixed but its 100% for heavier payloads.
- **KSC LC 39A:**
 - 100% success rate for payload ranges 2,000–5,500 kg and 9,000–12,500 kg.
 - 0% success rate for payloads between 5,500 kg and 7,000 kg.
 - 50% success rate at maximum payload capacities.

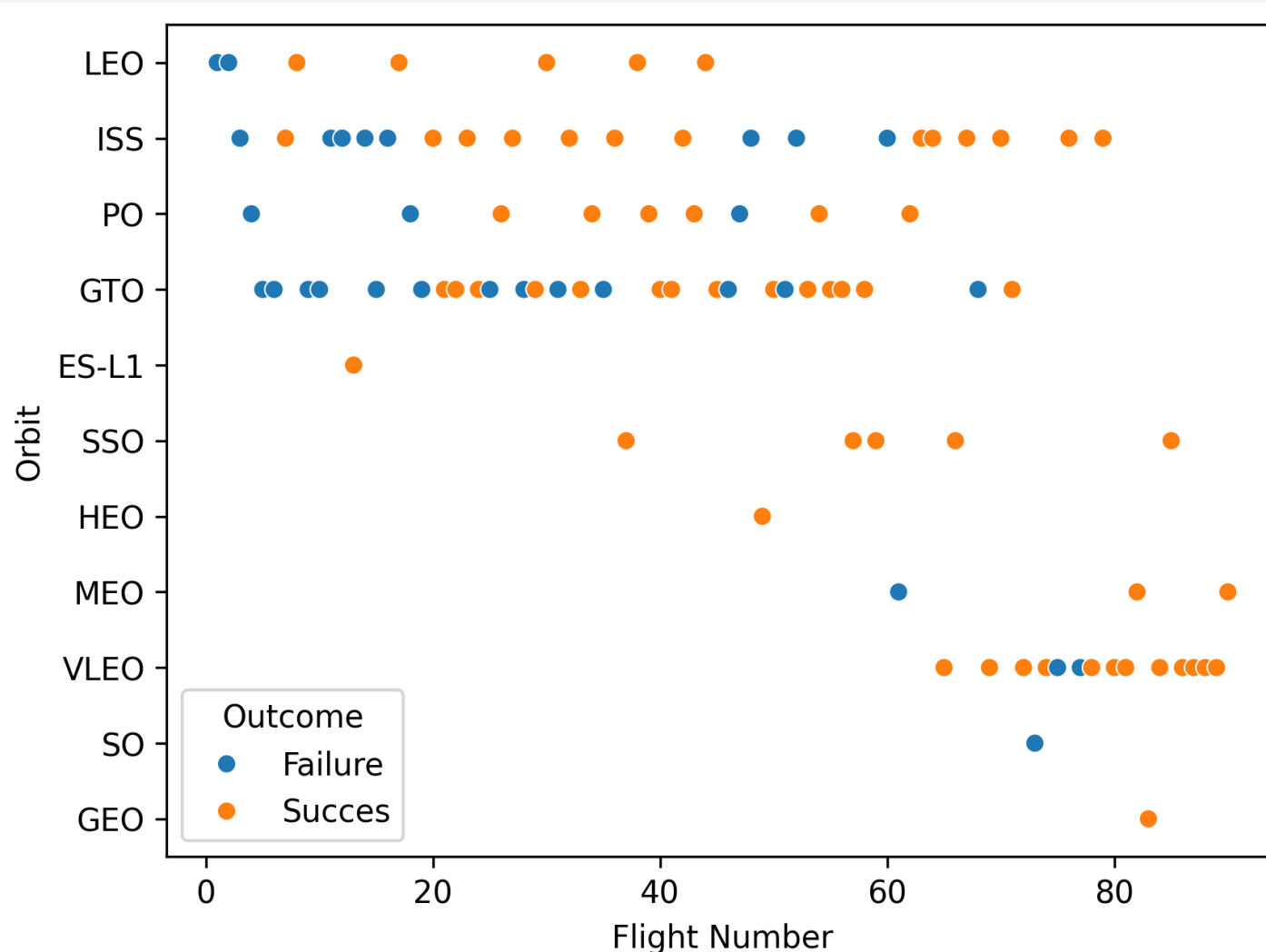
In conclusion payload capacity limits and landing outcomes vary significantly by launch complex due to specific orbital target requirements and site operational constraints.

Success Rate vs. Orbit Type



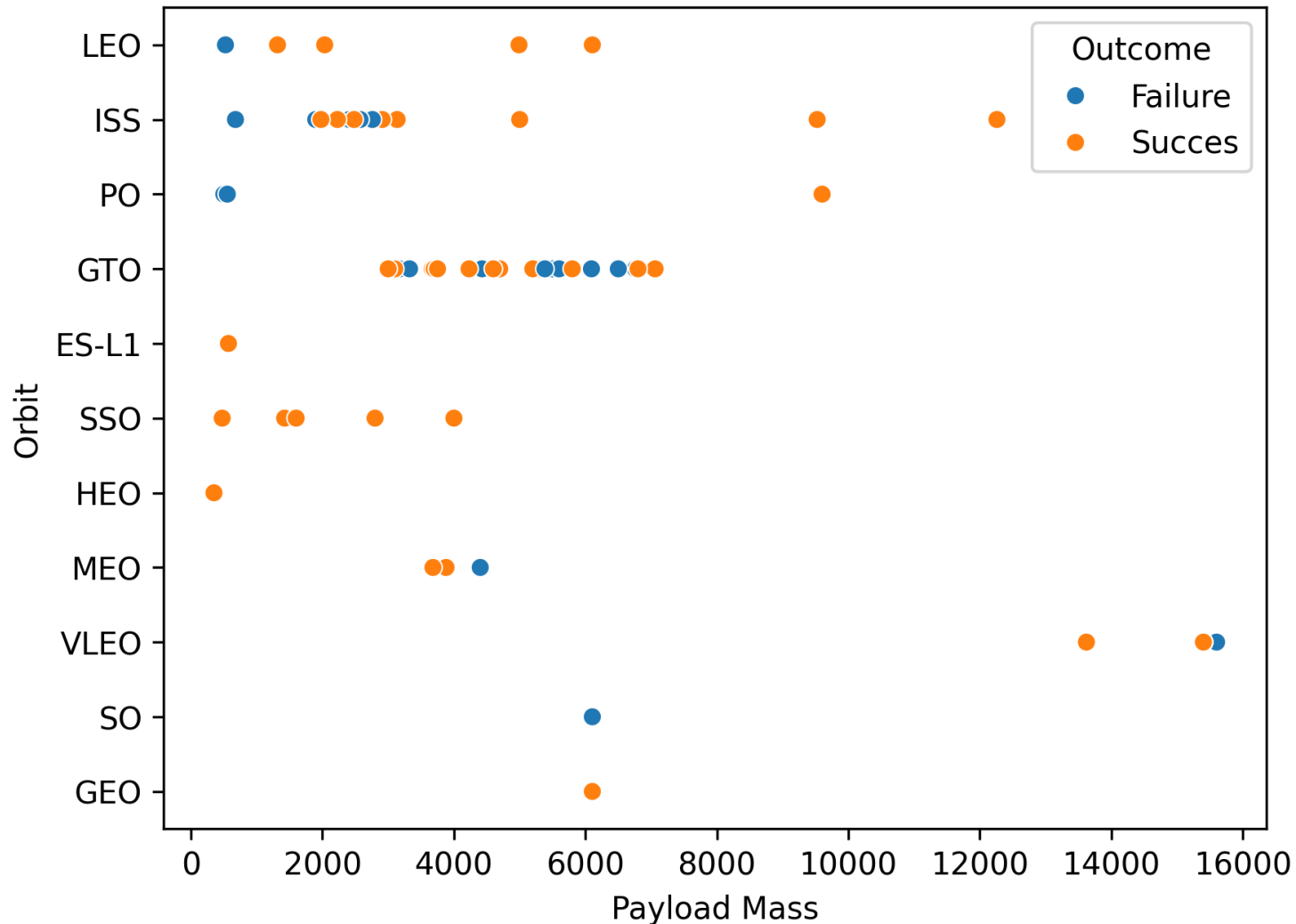
- **100% Success - ES-L1, GEO, HEO, SSO:** Flawless recovery across high-energy/sun-sync orbits
- **85% Success - VLEO:** High-reliability deployment profile.
- **60%–70% Success - ISS, LEO, MEO, PO:** Consistent core operational performance
- **50% Success - GTO:** High re-entry velocities & strict fuel constraints
- **0% Success - SO:** Single unsuccessful sub-orbital test flight

Flight Number vs. Orbit Type



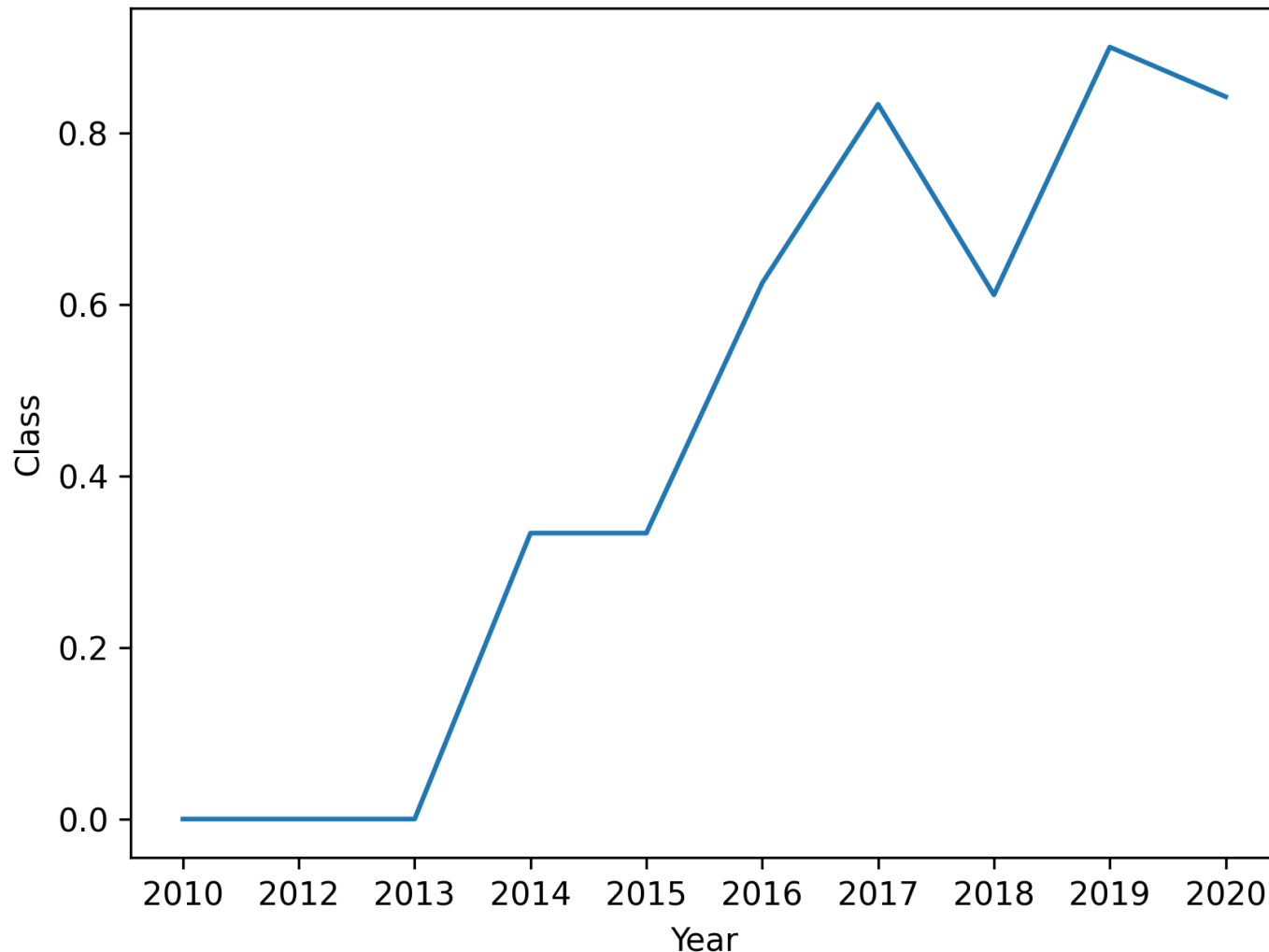
- **LEO, ISS, VLEO, MEO:** Success seems to be related to the number of flights.
- **GTO:** Has very mixed outcomes across all flight numbers so there appear to be no connection
- **ES-L1, SSO, HEO, GEO:** Have 100% success rate so no link could be observed

Payload vs. Orbit Type



- **LEO, ISS, PO:** There seems to be correlation between heavy payloads (>3500 kg) and landing success.
- **GTO:** As there are mixed landings for all weights there is no visible connection
- **VLEO:** Is used for heavy payloads (>12000) and has high success rate.

Launch Success Yearly Trend



- **2010–2013:** Near-zero landing success rate during early experimental phases.
- **2013–2015:** Steady upward trend with a temporary plateau in 2014 as technology improved
- **2015-2017:** Rapid growth and success rate exceeded 80%.
- **2017-2019:** There was a dip in 2018 where success rate fell to 65% but it resulted in increase in 2019 compared to 2017.
- **2019+:** slight downward trajectory but success rate over 80%

Overall the plot demonstrates a strong positive yearly correlation between experience and landing success.

All Launch Site Names

A query giving us all unique launch site names.

```
SELECT DISTINCT  
Launch_Site  
FROM SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

A query to display 5 records where launch sites begin with the string 'CCA'.*(Note. For simplicity sake only the Booster Version and the Launch Site results are given)*

SELECT *

FROM SPACEXTABLE

WHERE Launch_Site

LIKE 'CCA%'

LIMIT 5

Booster_Version	Launch_Site
F9 v1.0 B0003	CCAFS LC-40
F9 v1.0 B0004	CCAFS LC-40
F9 v1.0 B0005	CCAFS LC-40
F9 v1.0 B0006	CCAFS LC-40
F9 v1.0 B0007	CCAFS LC-40

Total Payload Mass

A query calculating the total payload mass carried by boosters launched by NASA (CRS)

```
SELECT  
SUM(PAYLOAD_MASS__KG_)  
FROM SPACEXTABLE  
WHERE CUSTOMER LIKE 'NASA%'
```

Total_Payload
99980

Average Payload Mass by F9 v1.1

A query calculating the average payload mass carried by booster version F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_)
      AS AVG_PAYLOAD_MASS
FROM SPACEXTABLE
WHERE Booster_Version
      LIKE 'F9 v1.1%'
```

AVG_PAYLOAD_MASS
2534.6666666666665

First Successful Ground Landing Date

A query calculating the first successful landing outcome on ground pad.

```
SELECT MIN(Date) AS  
First_success  
  
FROM SPACEXTABLE  
  
WHERE Landing_Outcome = 'Success  
(ground pad)'
```

First_success
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

A query listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
SELECT Payload
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success
(drone ship)'
      AND PAYLOAD_MASS__KG_ >=
4000
      AND Payload_Mass__Kg_ <=
6000
```

Payload
JCSAT-14
JCSAT-16
SES-10
SES-11 / EchoStar 105

Total Number of Successful and Failure Mission Outcomes

A query listing the total number of successful and failure mission outcomes

```
SELECT COUNT(), Mission_Outcome  
FROM SPACEXTABLE  
GROUP BY Mission_Outcome
```

COUNT()	Mission_Outcome
1	Failure (in flight)
98	Success
1	Success
1	Success (payload status unclear)

Boosters Carried Maximum Payload

A query listing all the booster versions that have carried the maximum payload mass.

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Payload_Mass__Kg_ = (
    SELECT MAX(Payload_Mass__Kg_)
    FROM
    SPACEXTABLE
)
```

Booster_Version

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

A query listing the records by month for the flights in 2015.

%%sql

```
SELECT SUBSTR(Date, 6, 2) AS  
Month, Landing_outcome,  
Booster_Version, Launch_Site
```

```
FROM SPACEXTABLE
```

```
WHERE SUBSTR(Date, 0, 5) =  
'2015'
```

```
AND Landing_Outcome =  
'Failure (drone ship)'
```

Month	Landing_Outcome	Booster_Version	Launch_Site
1	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
4	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

A query ranking the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order.

```
SELECT Landing_Outcome, COUNT()  
as Count
```

```
FROM SPACEXTABLE
```

```
WHERE DATE BETWEEN '2010-06-04'  
AND '2017-03-20'
```

```
GROUP BY Landing_Outcome
```

```
ORDER BY Count DESC
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

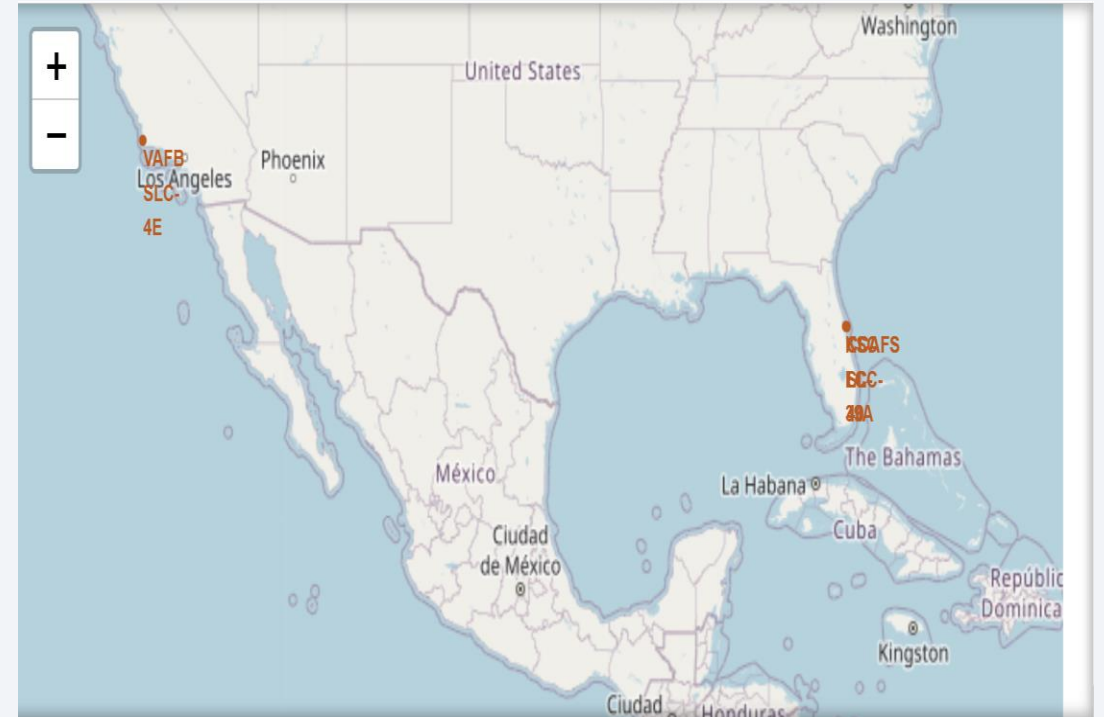
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

Section 3

Launch Sites Proximities Analysis

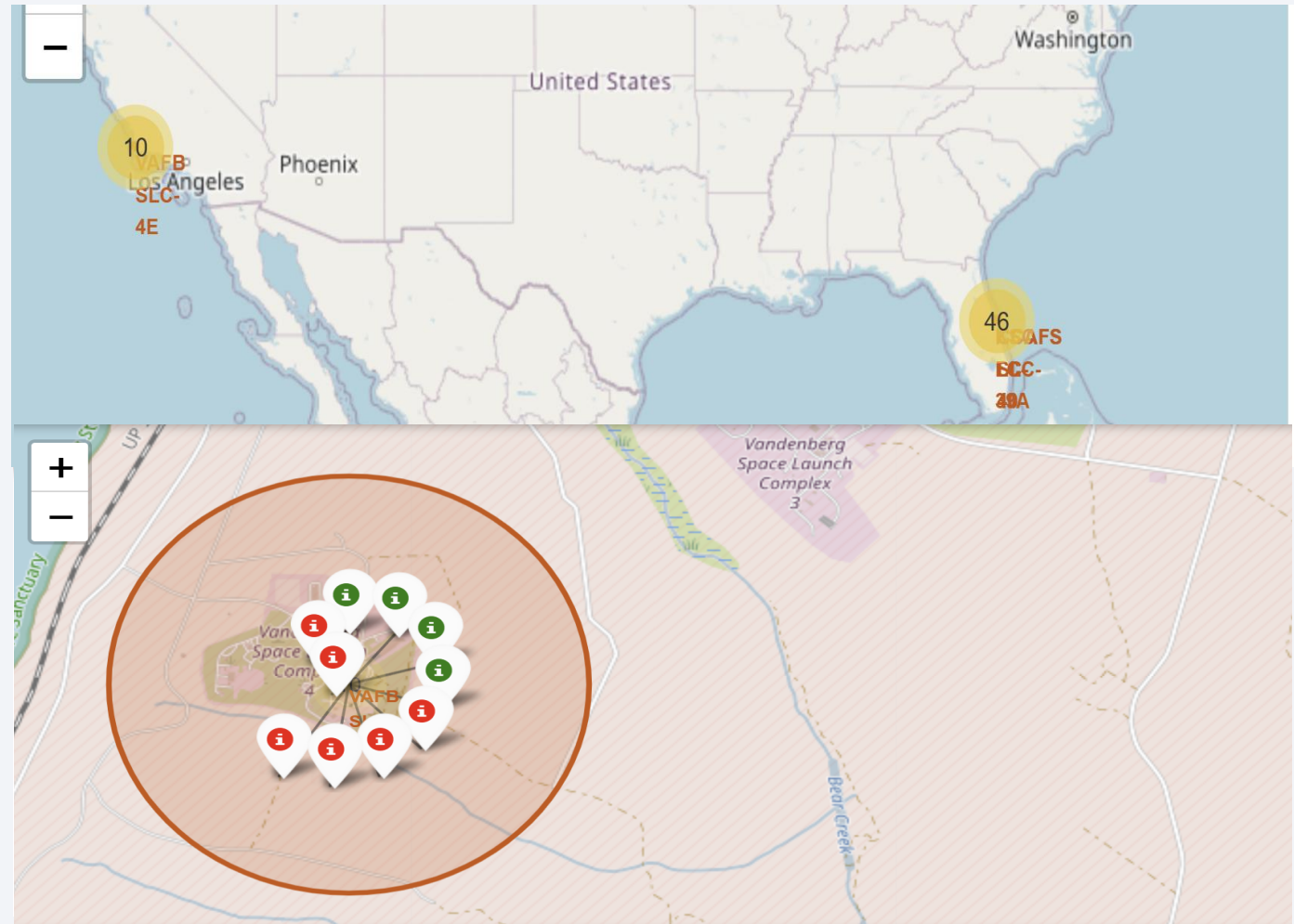
Launchpad site locations

- **Map Objects Used:** Marked all operational launch sites on the interactive map using **folium.Circle** and **folium.Marker**.
- **Southern Latitude:** All launch sites are situated in the southernmost regions of the United States to maximize Earth's rotational speed assistance during launch.
- **Dual-Coast Reach:** Maintaining active launch bases on both the East Coast and West Coast enables them to launch into low-inclination/equatorial orbits from the East and high-inclination/polar orbits from the West without overflying land.
- **Coastal Proximity:** Every site is located immediately adjacent to a coastline to enable downrange safety trajectories and sea-based booster recoveries.



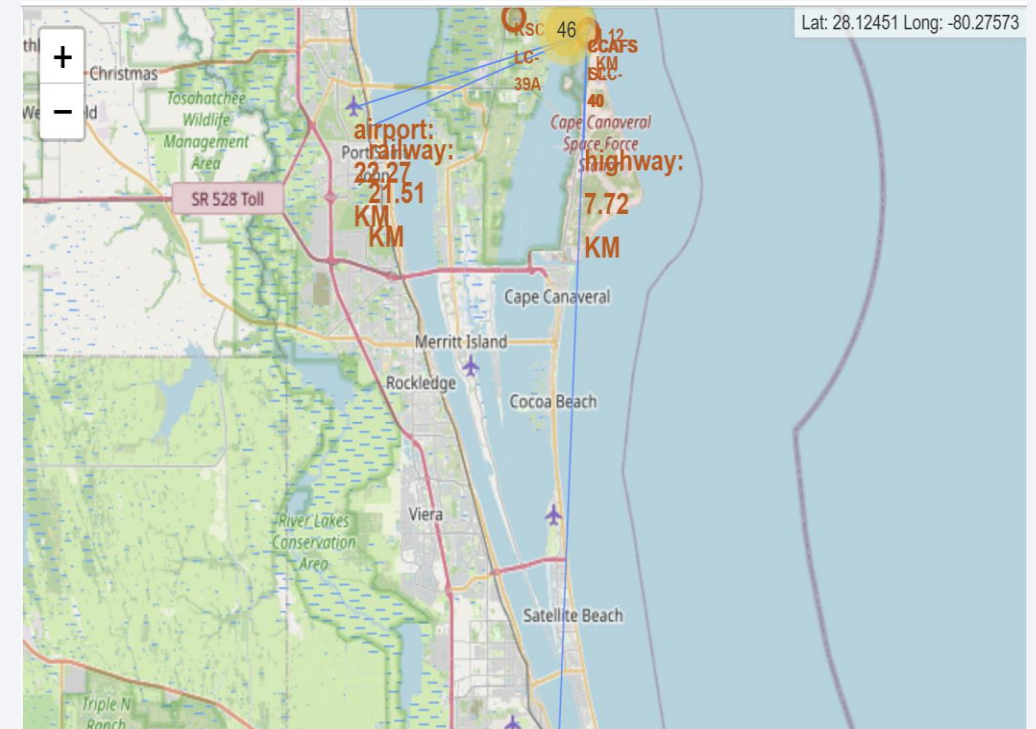
Launch Outcomes by Location

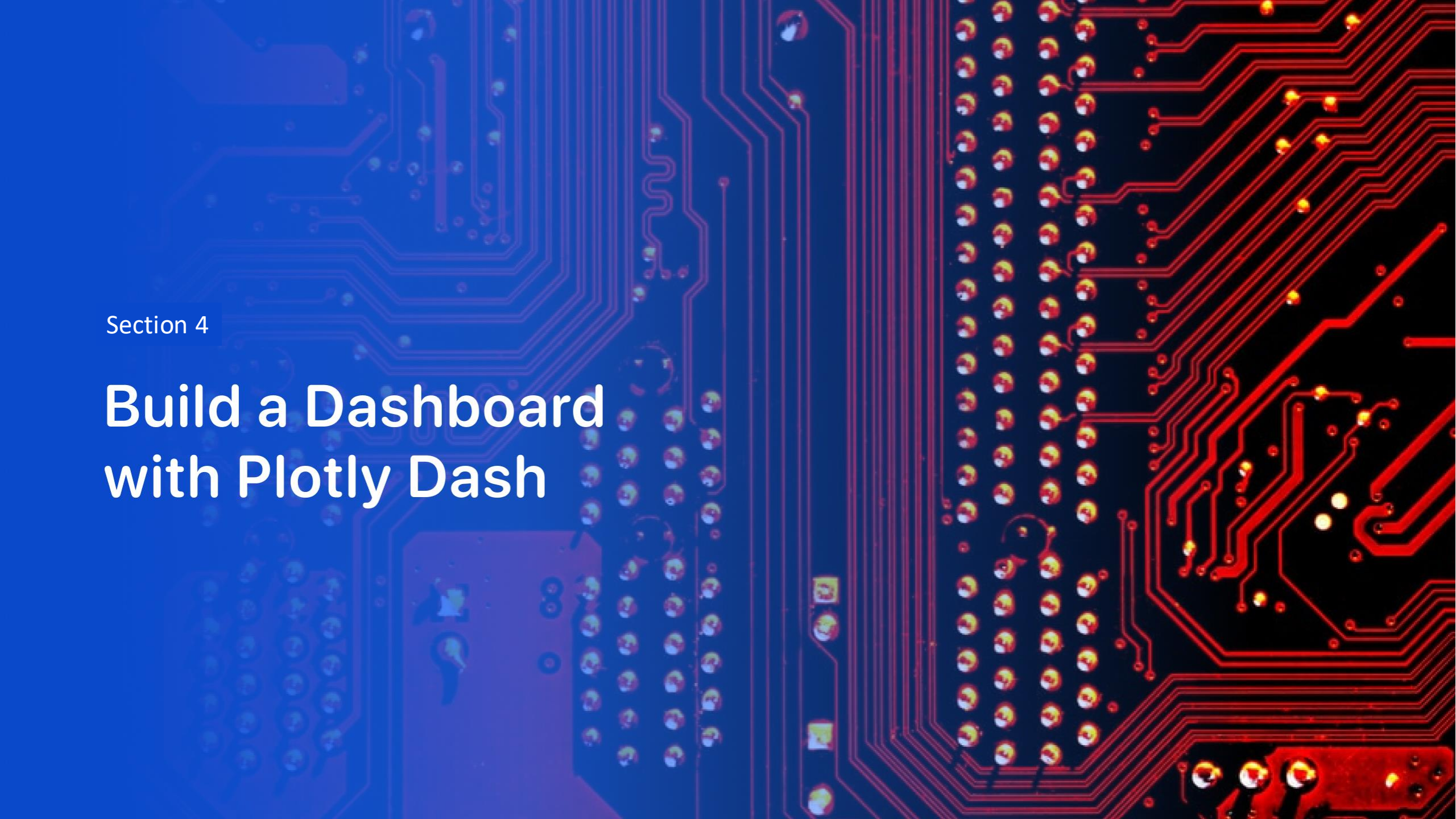
- **Map Objects Used:** Implemented `folium.plugins.MarkerCluster` with custom color-coded markers to evaluate site-specific recovery track records:
 - **green Markers (Class = 1):** Successful booster landings.
 - **red Markers (Class = 0):** Unsuccessful landing attempts or lost boosters.
- **Macro View (Coasts):** Before zooming in, marker clusters show total flights from the given coast
- **Micro View (Launch Pads):** Zooming into individual clusters reveals specific pad performance



Launch Site Proximity

- **Map Tools Used:** Combined `folium.plugins.MousePosition` to capture exact geographical coordinates, `folium.Marker` to pin specific points of interest and `folium.PolyLine` to draw direct distance lines
- **Proximity Features Measured:**
 - **Coastlines:** Minimize risks to populated areas during launches and facilitate ship landings.
 - **Railways & Highways:** Crucial for transporting oversized rocket stages, boosters, and heavy payload fairings directly to assembly facilities.
 - **Cities:** Positioned at a safe distance from residential areas to manage noise levels and launch hazard zones while remaining close enough to support workforces.



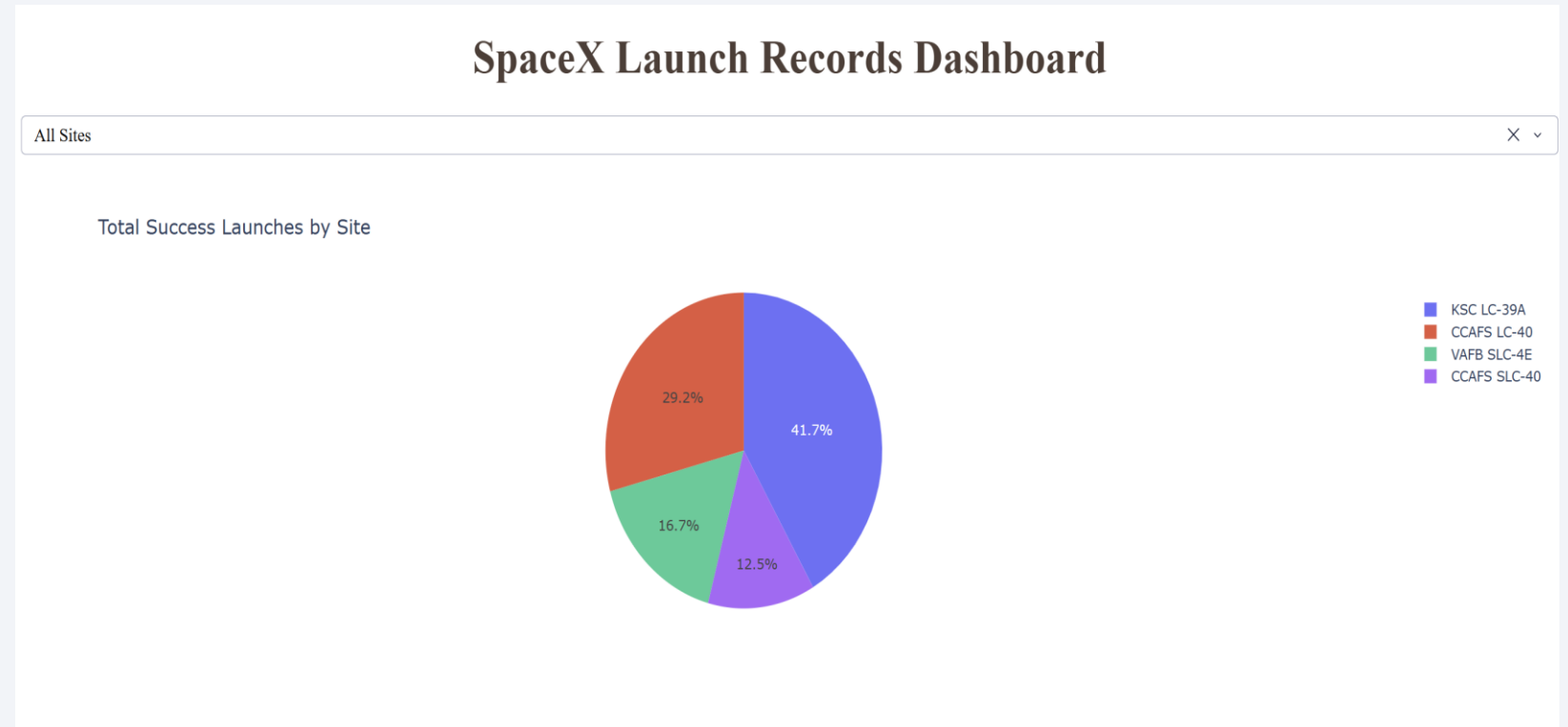


Section 4

Build a Dashboard with Plotly Dash

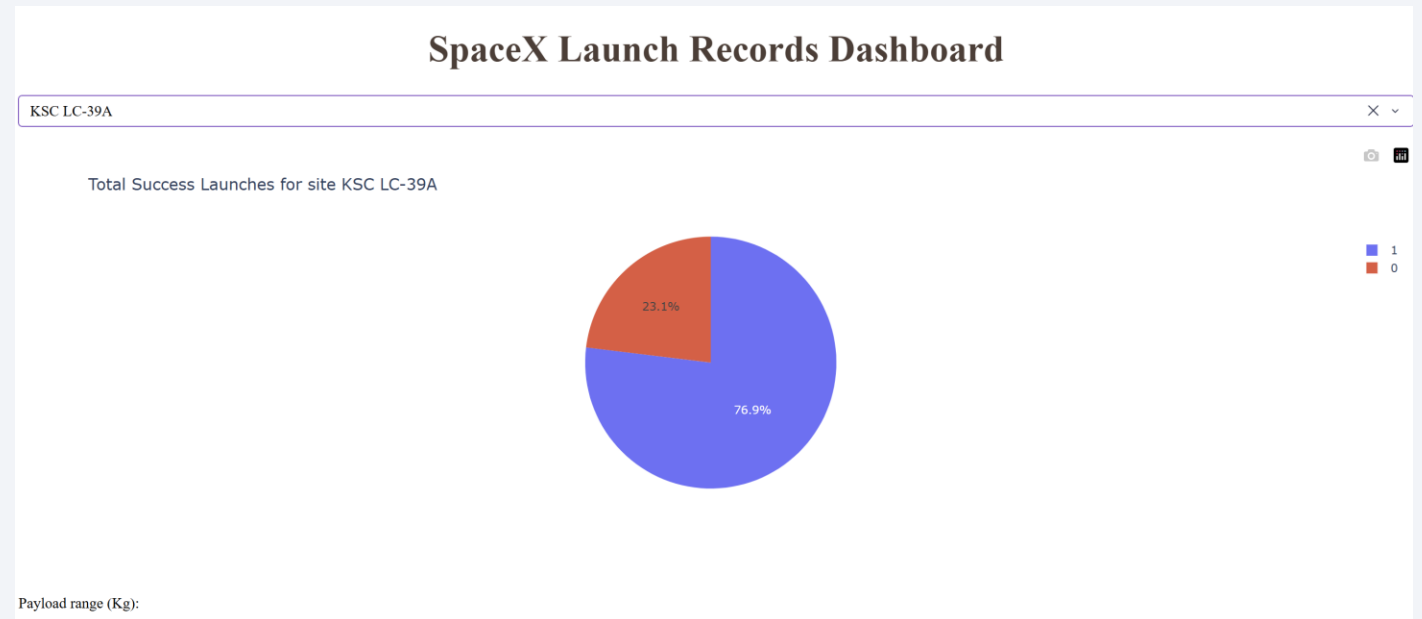
Total Successful Launches by Site

- **KSC LC-39A** has the highest proportion of successful landings at **41.7%** of all successful recoveries.
- **CCAFS LC-40** is the second highest at **29.2%**.
- **VAFB SLC-4E & CCAFS SLC-40** account for the remaining successful recoveries, with **CCAFS SLC-40** being the last.
- **KSC LC-39A** was the primary operational hub for successful landing outcomes.



KSC LC-39A - The most successful site

- The total successful launches for KSC LC-39A 76.9 % making it a highly reliable site.
- It handled high mission volume while maintaining a low failure rate.



Correlation between Payload Mass and Success for all Sites



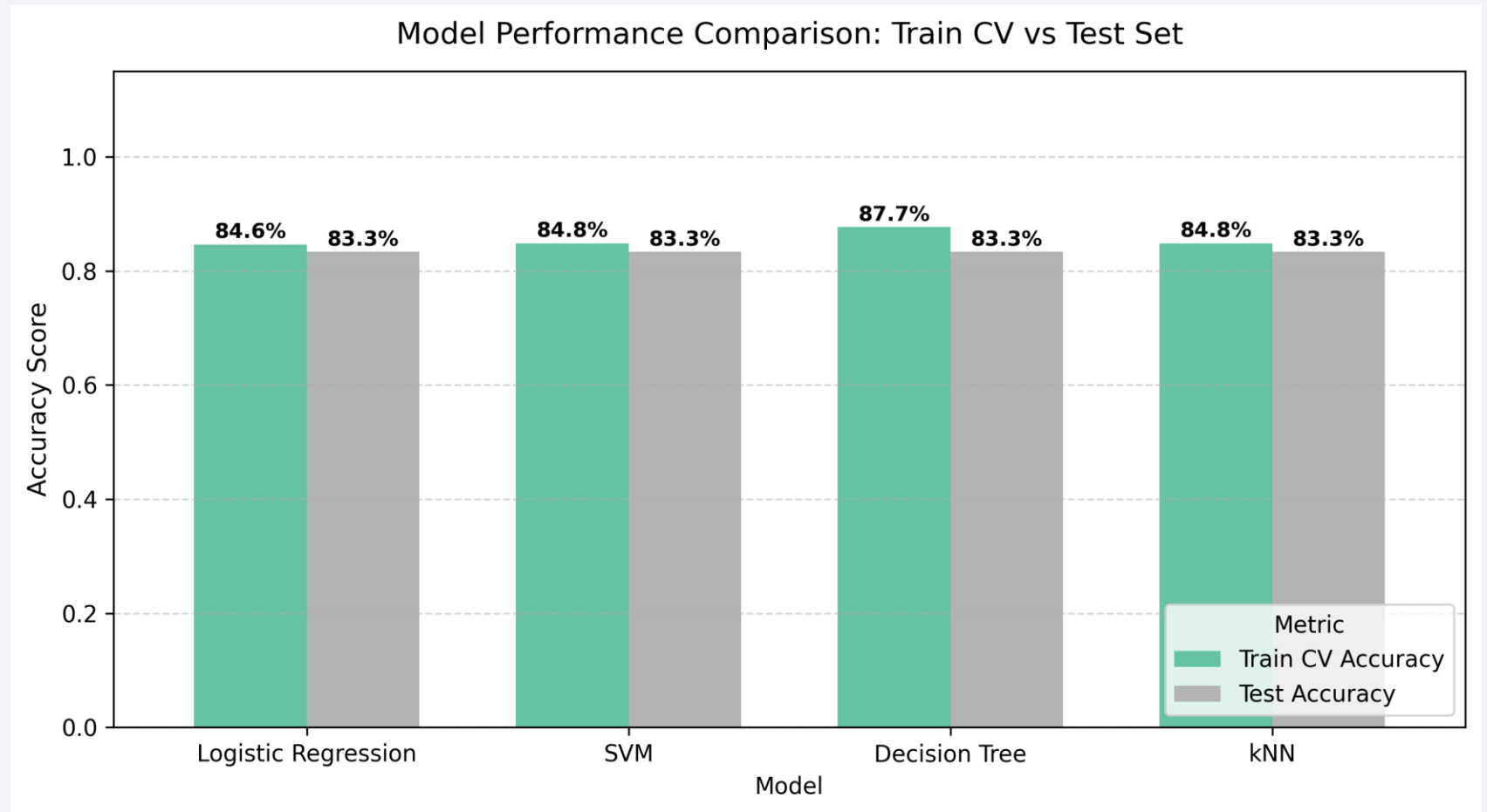
- **Booster Version Performance:**
 - **FT:** Most successful booster overall, achieving high recovery rate across multiple payload ranges.
 - **B4:** Second most successful booster, following FT's performance improvements.
 - **v1.0:** Recorded a **0% landing success rate**, reflecting early-stage experimental testing phases.
- **Payload Range Landing Outcomes:**
 - **5,000 kg – 6,000 kg:** FT boosters achieved a **50% success rate**.
 - **6,000 kg – 9,500 kg:** Experienced a **0% success rate** across attempts.
 - **9,600 kg:** Data is for B4, returning a **50% success rate**.

Section 5

Predictive Analysis (Classification)

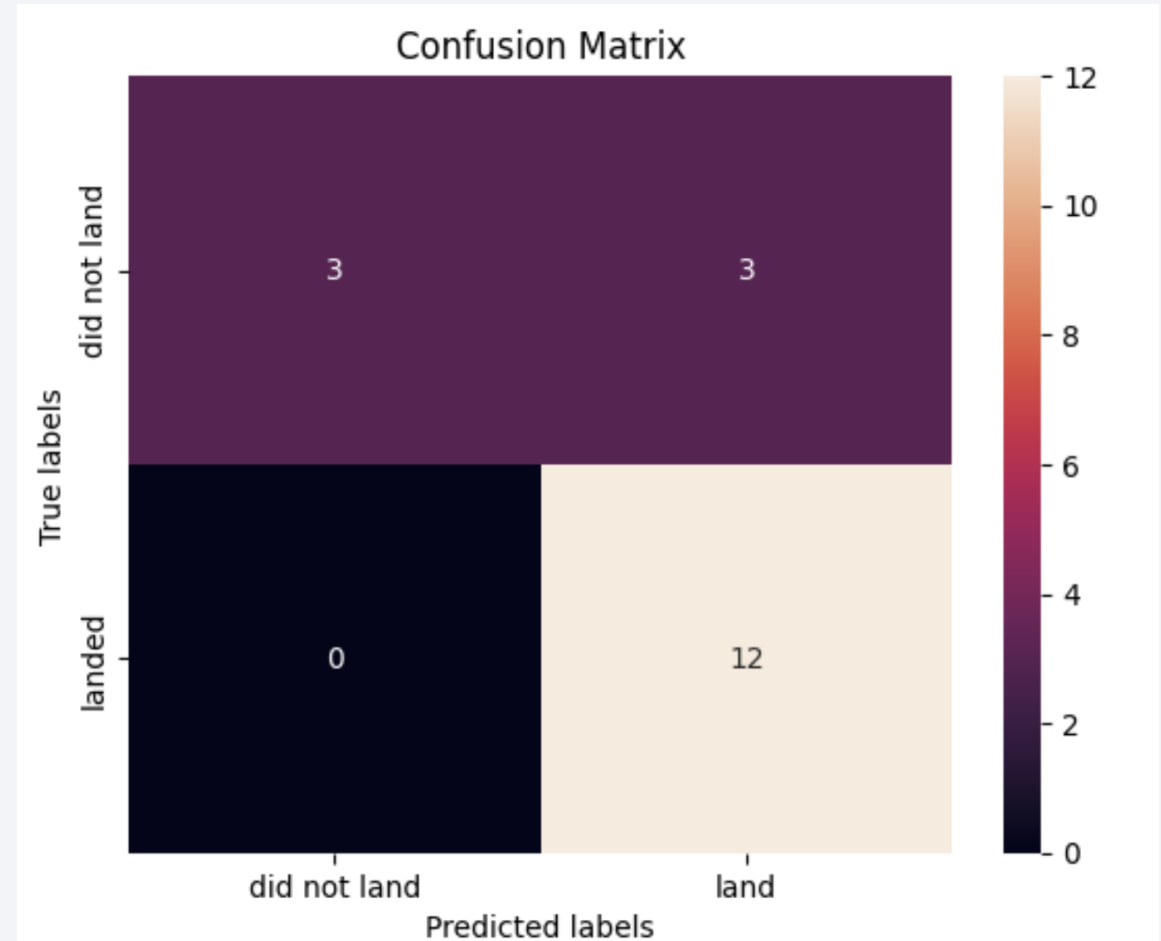
Classification Accuracy

- **All models** have the same **83.33%** accuracy on test data(N=18)
- **Decision Tree:** Reached the highest cross-validation score (~87.5%), but showed high variance across data splits.
- **kNN:** Showed stable, reproducible cross-validation scores (~84.6%) with a minimal gap between train and test performance.



Confusion Matrix

- **Zero False Negatives:** The model correctly identified 100% of all successful landing attempts without missing a single success.
- **3 False Positives:** The model misclassified 3 actual landing failures as successful landings.
- The confusion matrix shows a mild bias toward predicting success, driven by the historical imbalance of successful launches in recent years.



Conclusions

- **Exploratory Data Analysis (EDA)**

- **Success Trend Over Time:** Strong positive correlation between operational experience and landing success—progressing from 0% in early testing (2010–2013) to a sustained success rate above 80% (2015+).
- **Booster Evolution:** Clear reliability jump across hardware iterations, moving from early v1.0 failures to high recovery counts on FT and Block 4 boosters.
- **Core Takeaway:** Payload limits, orbital targets (ES-L1/GEO at 100% vs. GTO at 50%), and landing outcomes vary significantly by launch complex due to specific operational constraints.

Conclusions

- **Geographic & Proximity Insights (Folium):**
 - Launch sites are strategically placed in southern latitudes along coastlines(maximizing rotational velocity boost and trajectory safety).
 - Pads are positioned adjacent to critical transport infrastructure (rail/highways)for heavy booster logistics, while maintaining safe buffer distances from majorpopulation centers.

Conclusions

- **Interactive Dashboard Analysis (Dash):**
 - **Pad Reliability (KSC LC-39A):** Proved to be the most dependable launch complex, maintaining a **76.9% site success rate** and contributing **41.7% of all successful fleet landings** while handling high mission volume.
 - **Booster & Payload Performance:** Operational success was driven by **FT** and **B4** boosters (replacing the 0% success rate of early v1.0 variants), with 50% success rates in the 5,000–6,000 kg (FT) and 9,600 kg (B4) ranges, but a 0% success rate in the heavy 6,000–9,500 kg payload band.
- **Predictive Modeling (Machine Learning):**
 - All classification models achieved 83.33% Test Accuracy ($N=18$), with 0 False Negatives on successful recoveries. k-Nearest Neighbors (KNN) was selected as the optimal deployment model due to its low variance and consistent cross-validation stability (~84.6%).

Appendix

- EDA
- # Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
 - `g = sns.catplot(y='LaunchSite', x='FlightNumber', data=df, hue='Class', height=5, aspect=2)`
 - `g._legend.set_title("Landing Outcome")`
 - `new_labels = ['Failure', 'Success']`
 - `for t, l in zip(g._legend.texts, new_labels):`
 - `t.set_text(l)`
 - `plt.xlabel("Flight Number")`
 - `plt.ylabel('Launch Site')`
 - `plt.savefig("plots/flight_vs_launchsite.png", dpi=300, bbox_inches='tight')`
 - `plt.show()`
- # Plot a line chart with x axis to be the extracted year and y axis to be the success rate
 - `yearly_success_rate = df.groupby('Date', as_index=False)['Class'].mean()`
 - `sns.lineplot(data=yearly_success_rate, x='Date', y='Class')`
 - `plt.xlabel('Year')`
 - `plt.savefig("plots/success_rate_over_years.png", dpi=300, bbox_inches='tight')`
 - `plt.show()`

Appendix

- SQL
- %%sql
- SELECT Booster_Version
- FROM SPACEXTABLE
- WHERE Payload_Mass__Kg_ = (
 - SELECT MAX(Payload_Mass__Kg_)
 - FROM SPACEXTABLE
 -)
- %%sql
- SELECT SUBSTR(Date, 6, 2) AS Month, Landing_outcome, Booster_Version, Launch_Site
- FROM SPACEXTABLE
- WHERE SUBSTR(Date, 0, 5) = '2015'
- AND Landing_Outcome = 'Failure (drone ship)'

Appendix

- Folium
- for type_com, coords in close_comunications.items():
 - dist_curr = calculate_distance(launch_site_lat, launch_site_lon, coords[0], coords[1])
 - distance_marker = folium.Marker(
 - coords,
 - icon=DivIcon(
 - icon_size=(20,20),
 - icon_anchor=(0,0),
 - html=f'<div style="font-size: 12pt; color:#d35400;">{type_com}: {dist_curr:.2f} KM</div>')
 -)
 - site_map.add_child(distance_marker)
- # Create a `folium.PolyLine` object using the coastline coordinates and launch site coordinate
- lines=folium.PolyLine(locations=[[launch_site_lat, launch_site_lon], coords], weight=1)
- site_map.add_child(lines)

Appendix

- Machine Learning
- `accuracy_df = pd.DataFrame(model_accuracy)`
- `accuracy_df['Train CV Accuracy (%)'] = (accuracy_df['Train CV Accuracy'] * 100).round(2)`
- `accuracy_df['Test Accuracy (%)'] = (accuracy_df['Test Accuracy'] * 100).round(2)`
- # Display the summary table
- `accuracy_df[['Model', 'Train CV Accuracy (%)', 'Test Accuracy (%)']]`

Thank you!

